UNIT IV

Queuing Theory & Inventory Control

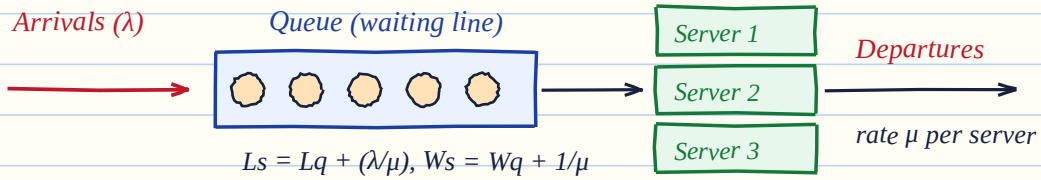
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Syllabus topics covered in this unit:

- Queuing: terminology, arrival and service patterns, birth-death process
- Single channel models (M/M/1) with infinite and finite queue length
- Multichannel model (M/M/C) with infinite queue length
- Inventory: EOQ with and without shortages, production model
- Buffer stock, quantity discount (single price break), selective control

1. Queuing Theory - Introduction

A **queue (waiting line)** forms when customers arriving for service must wait because the service facility is busy. Queuing theory builds mathematical models of arrivals, waiting and service in order to **balance the cost of waiting against the cost of providing service**. Examples: bank counters, hospital OPD, toll gates, ATM, airport runway, machines waiting for repair.



Elements / terminology of a queuing system

- *Input (arrival) pattern:* how customers arrive - usually **Poisson** with mean rate λ per unit time, i.e. inter-arrival times are exponential. May be single or bulk, finite or infinite
- *Service channel (mechanism):* number of servers and their arrangement - single channel, multi-channel in parallel, or in series (phases)
- *Service (departure) pattern:* time taken to serve a customer - usually **exponential** with mean service rate μ per unit time (mean service time 1/μ)
- *Population (calling source):* infinite (large) or finite source of customers
- *Queue discipline:* rule for selecting the next customer - **FIFO/FCFS**, LIFO, SIRO (random), priority (PR), pre-emptive priority
- *Customer behaviour:* balking (refuses to join a long queue), reneging (leaves after waiting), jockeying (switches queues), collusion
- *Capacity of system:* infinite or finite (maximum N in system); *steady state:* probabilities do not change with time (needs $\lambda < c\mu$)

Notation

Symbol	Meaning	Symbol	Meaning
λ	mean arrival rate	ρ	traffic intensity λ/μ (or $\lambda/c\mu$)
μ	mean service rate	P_n	probability of n in system
L_s	mean number in system	L_q	mean number in queue
W_s	mean time in system	W_q	mean waiting time in queue
c	number of servers	N	system capacity

Kendall's notation: $(a / b / c) : (d / e)$ a = arrival distribution, b = service distribution (M = Markovian/Poisson-exponential, D = deterministic, G = general), c = number of servers, d = system capacity, e = queue discipline. Example $(M/M/1):(\infty/FIFO)$.

Little's formulae: $L_s = \lambda W_s$, $L_q = \lambda W_q$, $W_s = W_q + 1/\mu$, $L_s = L_q + \lambda/\mu$.

2. Birth and Death Process

Arrival = **birth** (system size n -> n+1 with rate λ_n); departure = **death** (n -> n-1 with rate μ_n). In steady state the rate of flow **into** a state equals the rate **out of** it (balance equations).

State 0: $\lambda_0 P_0 = \mu_1 P_1$

State n ($n \geq 1$): $\lambda_{n-1} P_{n-1} + \mu_{n+1} P_{n+1} = (\lambda_n + \mu_n) P_n$

Solving: $P_n = [\, \lambda_0 \lambda_1 \dots \lambda_{n-1} / (\mu_1 \mu_2 \dots \mu_n) \,] P_0$, and $\sum P_n = 1$ gives P_0.

For constant rates $\lambda_n = \lambda$, $\mu_n = \mu$: $P_n = (\lambda/\mu)^n P_0 = \rho^n P_0$

Summing the geometric series: $P_0 = 1 - \rho$ (for $\rho < 1$) hence $P_n = (1 - \rho)\rho^n$. All the standard queue models are derived from this.

3. Model I: (M/M/1):(∞/FIFO) - Single Server, Infinite Queue

Conditions: Poisson arrivals (λ), exponential service (μ), one server, infinite capacity, FIFO, $\rho = \lambda/\mu < 1$.

$P_0 = 1 - \rho$; $P_n = (1 - \rho)\rho^n$; $P(n \geq k) = \rho^k$

$L_s = \rho/(1 - \rho) = \lambda/(\mu - \lambda)$ $L_q = \rho^2/(1 - \rho) = \lambda^2/[\mu(\mu - \lambda)]$

$W_s = 1/(\mu - \lambda)$ $W_q = \lambda/[\mu(\mu - \lambda)]$ $P(W_s > t) = e^{-(\mu-\lambda)t}$ L_n (non-empty queue) $= \mu/(\mu - \lambda)$

Example 1

Customers arrive at a service counter at 6 per hour (Poisson); service is exponential at 8 per hour. Find the performance measures.

Measure	Formula	Value
Traffic intensity ρ	λ/μ	$6/8 = 0.75$
P(system empty) P_0	$1 - \rho$	0.25
Mean number in system L_s	$\lambda/(\mu-\lambda)$	6/2 = 3 customers
Mean queue length L_q	$\rho^2/(1-\rho)$	0.5625/0.25 = 2.25
Time in system W_s	$1/(\mu-\lambda)$	0.5 hr = 30 min
Waiting time W_q	$\lambda/[\mu(\mu-\lambda)]$	6/16 = 0.375 hr = 22.5 min
P(3 or more in system)	ρ^3	0.4219
P(time in system > 30 min)	$e^{-(\mu-\lambda)(0.5)}$	$e^{-1} = 0.368$

Check by Little's law: $L_s = \lambda W_s = 6 \times 0.5 = 3$ and $L_q = \lambda W_q = 6 \times 0.375 = 2.25$ (agrees). The server is busy 75% of the time ($= \rho$).

Derivation of L_s and L_q

$L_s = \sum n P_n = (1 - \rho) \sum n \rho^n = (1 - \rho)\, \rho/(1 - \rho)^2 = \rho/(1 - \rho)$ (using $\sum n\rho^n = \rho/(1-\rho)^2$).

$L_q = \sum_{n\geq 1} (n - 1) P_n = L_s - (1 - P_0) = \rho/(1-\rho) - \rho = \rho^2/(1 - \rho)$.

Waiting time in system is exponential with mean $1/(\mu - \lambda)$, hence $W_s = L_s/\lambda = 1/(\mu - \lambda)$.

Example 1(b): design question

$\lambda = 10$ per hour, $\mu = 12$ per hour: $\rho = 0.833$, $L_s = 5$, $L_q = 4.17$, $W_s = 0.5$ hr, $W_q = 0.417$ hr.

What μ is needed so that the mean time in the system is at most 15 min? $W_s = 1/(\mu - 10) \leq 0.25$ hr $\Rightarrow$ $\mu - 10 \geq 4$ $\Rightarrow$ $\mu \geq 14$ per hour.

4. Model II: (M/M/1):(N/FIFO) - Finite Queue Length

At most **N customers** can be in the system (including the one in service); an arrival finding N present is turned away (**lost**). The system is stable for any ρ, even $\rho \geq 1$.

$P_0 = (1 - \rho)/(1 - \rho^{N+1})$ and $P_n = (1 - \rho)\, \rho^n/(1 - \rho^{N+1})$, $n = 0, 1, \ldots, N$ ($\rho \neq 1$; if $\rho = 1$, $P_n = 1/(N+1)$)

$L_s = \rho/(1-\rho) - (N+1)\rho^{N+1}/(1 - \rho^{N+1})$ $\quad L_q = L_s - (1 - P_0)$

Effective arrival rate $\lambda' = \lambda(1 - P_N)$ (P_N = probability an arrival is lost)

$W_s = L_s/\lambda'$, $\quad W_q = L_q/\lambda' = W_s - 1/\mu$

Example 2

$\lambda = 3$ per hour, $\mu = 4$ per hour and the room holds at most $N = 4$ customers. $\rho = 0.75$, $\rho^5 = 0.2373$.

Quantity	Working	Result
P_0	(1-0.75)/(1-0.2373)	0.3278
$P_4 = P(\text{lost})$	0.3278×0.75^4	0.1037 (10.4%)
L_s	3 - 5(0.2373)/0.7627	1.444
Effective λ'	3(1 - 0.1037)	2.689 per hr
L_q	1.444 - (1 - 0.3278)	0.772
W_s	1.444/2.689	0.537 hr (32.2 min)
W_q	0.772/2.689	0.287 hr (17.2 min)

5. Model III: (M/M/C):(∞/FIFO) - Multichannel

c identical servers in parallel, one common queue, each server has rate μ. Overall service rate = $n\mu$ when $n \le c$ busy, and $c\mu$ when $n \ge c$. Steady state needs $\rho = \lambda/(c\mu) < 1$.

$$P_0 = \left[\sum_{n=0}^{c-1} (\lambda/\mu)^n/n! \; + \; (\lambda/\mu)^c / (c! \, (1 - \lambda/c\mu)) \right]^{-1}$$

$$P_n = (\lambda/\mu)^n P_0/n! \;\; (n < c); \quad P_n = (\lambda/\mu)^n P_0/(c! \, c^{n-c}) \;\; (n \ge c)$$

$$L_q = P_0 (\lambda/\mu)^c \rho / [c! \, (1 - \rho)^2], \quad L_s = L_q + \lambda/\mu$$

$$W_q = L_q/\lambda, \quad W_s = W_q + 1/\mu = L_s/\lambda. \quad P(\text{wait}) = (\lambda/\mu)^c P_0 / [c!(1 - \rho)]$$

Example 3

Arrival rate $\lambda = 12$ per hour, each of $c = 2$ servers serves $\mu = 8$ per hour. Then $\lambda/\mu = 1.5$ and $\rho = 12/16 = 0.75$.

Quantity	Working	Result
P_0	$[1 + 1.5 + (1.5)^2/(2 \times 0.25)]^{-1} = [1 + 1.5 + 4.5]^{-1}$	1/7 = 0.1429
P(customer waits)	(2.25/(2×0.25)) × (1/7) = 4.5/7	0.643
L_q	(1/7)(2.25)(0.75)/[2(0.0625)]	1.929
L_s	1.929 + 1.5	3.429
W_q	1.929/12	0.161 hr = 9.6 min
W_s	0.161 + 1/8	0.286 hr = 17.1 min

Example 3(b): three servers

$\lambda = 18$ per hour, $\mu = 8$ per hour per server, $c = 3$: $\lambda/\mu = 2.25$, $\rho = 0.75$. $P_0 = [1 + 2.25 + 2.531 + 7.594]^{-1} = 1/13.375 = 0.0748$.

L_q	L_s	W_q	W_s
1.703	3.953	0.0946 hr (5.7 min)	0.2196 hr (13.2 min)

Compare Example 3 (2 servers, $\lambda = 12$): adding a server to handle more load keeps $\rho = 0.75$ but

a pooled queue gives shorter waits than separate single-server queues of the same load.

Summary of the three models

Model	Notation	Key formula
I	(M/M/1):(∞/FIFO)	$L_s = \lambda/(\mu-\lambda)$
II	(M/M/1):(N/FIFO)	$P_n = (1-\rho)\rho^n/(1-\rho^{N+1})$; $\lambda' = \lambda(1-P_N)$
III	(M/M/c):(∞/FIFO)	$L_q = P_0(\lambda/\mu)^c\rho/[c!(1-\rho)^2]$

Cost model: total cost per hour = (cost per server × c) + (waiting cost per customer-hour × L_s). Choose c that minimises it.

6. Inventory Control - Introduction

Inventory is the stock of materials (raw material, work-in-process, finished goods, spares) held to meet future demand. Inventory control decides **how much to order and when**, so that total cost is minimum while service is not interrupted.

Why hold inventory?

- To avoid stock-outs, to take advantage of quantity discounts, to smooth production, to guard against price rises and delays in delivery

Inventory costs

Cost	Symbol	Meaning
Holding (carrying) cost	C_h or C_1	per unit per year: interest, storage, insurance, obsolescence (often 15-30% of price)
Ordering (set-up) cost	C_0 or C_3	per order / per production run: paper work, transport, set-up
Shortage cost	C_s or C_2	per unit short per year: lost sales, penalty, goodwill
Purchase (production) cost	P	price per unit; matters only if price varies with Q

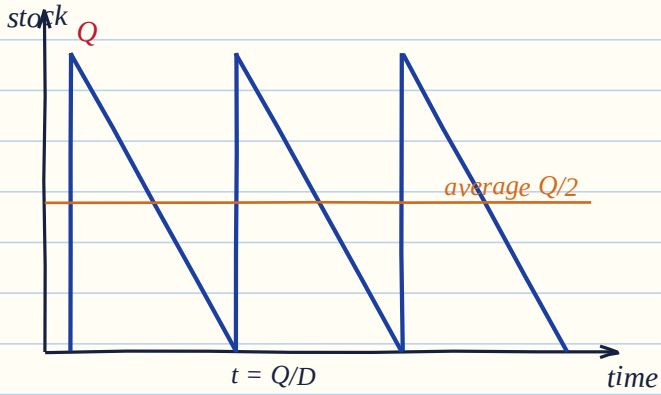
Terms

- D = annual demand; Q = order (lot) size; t = cycle time = Q/D; lead time L = delay between order and receipt
- Reorder level = demand during lead time + buffer stock; Buffer (safety) stock = protection against variation in demand/lead time

Deterministic models assume demand D and lead time are known and constant. Total annual cost = purchase + ordering + holding (+ shortage).

7. EOQ Model I - Economic Order Quantity (no shortages)

Assumptions: constant known demand rate, ***instantaneous replenishment****, no shortages, lead time zero/constant, price constant.*

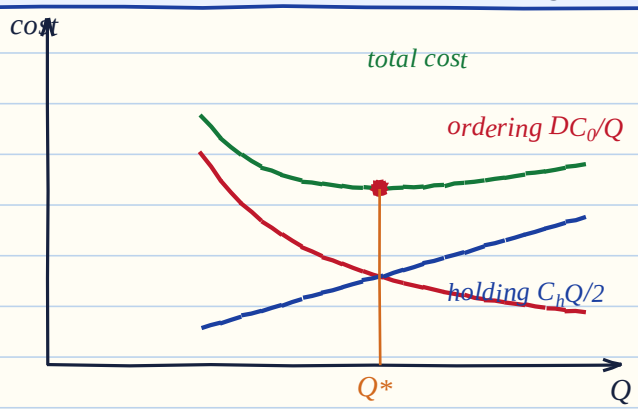


Annual ordering cost = $(D/Q)\,C_0$; annual holding cost = $(Q/2)\,C_h$. $TC(Q) = DP + DC_0/Q + C_hQ/2$

$dTC/dQ = 0$ gives $Q^* = \sqrt{(2DC_0/C_h)}$, $TC_{min} = DP + \sqrt{(2DC_0C_h)}$

Number of orders $N^* = D/Q^*$; cycle time $t^* = Q^*/D$. At the optimum ***ordering cost = holding cost****.*

Second derivative $= 2DC_0/Q^3 > 0$, so Q^* gives a minimum.



Example 4

$D = 2400$ units/year, $C_0 = Rs\ 100$ per order, $C_h = Rs\ 6$ per unit per year.

$Q^* = \sqrt{(2 \times 2400 \times 100/6)} = \sqrt{80000} = 283$ units (282.84)

Annual cost (excluding purchase) $= \sqrt{(2 \times 2400 \times 100 \times 6)} = \sqrt{2880000} = Rs\ 1697$ (ordering 848.5 + holding 848.5)

Orders per year $= 2400/282.84 = 8.49$; cycle time $= 0.1179$ yr ≈ 43 days.

Example 4(b): with purchase price

$D = 9000$ units/yr, price Rs 20, holding cost 20% of price ($C_h = Rs\ 4$), $C_0 = Rs\ 150$. $Q^* = \sqrt{(2 \times 9000 \times 150/4)} = \sqrt{675000} = 821.6$. Orders/yr = 10.95; cycle = 33 days. Annual cost $= 9000 \times 20 + \sqrt{(2 \times 9000 \times 150 \times 4)} = 180000 + 3286.3 = Rs\ 1,83,286$.

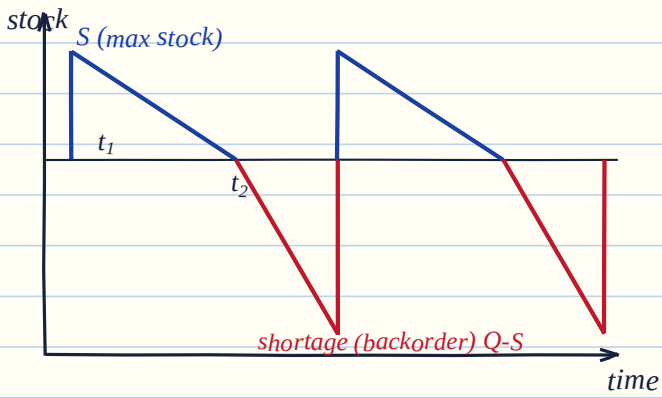
Assumptions and limitations of EOQ

- Demand constant and known; price independent of Q; whole lot arrives together; no stock-outs; single item, no capacity limit
- TC curve is **flat** near Q*: a 20% error in Q changes TC by only about 2% - EOQ is robust

Model	Demand	Replenishment	Shortages
EOQ I	constant	instantaneous	not allowed
EOQ II	constant	instantaneous	back-ordered
Production	constant	finite rate K	not allowed
Probabilistic	random	any	buffer stock used

8. EOQ Model II - With Shortages (back-ordering)

Shortages are allowed and **fully back-ordered**; the backlog is filled as soon as the next lot arrives. Shortage cost C_s per unit per year.



$Q^* = \sqrt{[\ 2DC_0(C_h + C_s) / (C_h\ C_s)\ ]} = EOQ \times \sqrt{[(C_h + C_s)/C_s]}$

Maximum inventory $S^* = Q^*\ C_s/(C_h + C_s)$, maximum shortage $Q^* - S^* = Q^*\ C_h/(C_h + C_s)$

$TC_{min} = \sqrt{[\ 2DC_0C_hC_s/(C_h + C_s)\ ]}$ $t_1 = S^*/D$ (stock period), $t_2 = (Q^*-S^*)/D$ (shortage period)

Example 5: same data as Example 4 with C_s = Rs 12 per unit per year.

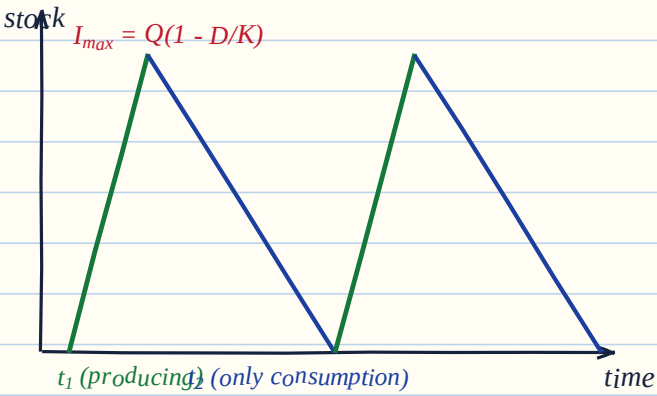
$Q^* = \sqrt{(2 \times 2400 \times 100 \times 18/72)} = \sqrt{120000} = 346.4$ units; $S^* = 346.4 \times 12/18 = 230.9$; maximum shortage = 115.5

$TC = \sqrt{(2 \times 2400 \times 100 \times 6 \times 12/18)} =$ Rs 1385.6 < 1697 (cheaper than the no-shortage plan).

As $C_s \to \infty$ the formula reduces to Model I (no shortages are worth allowing).

9. Production (Manufacturing) Model - finite replenishment

Items are produced in lots at rate K per year ($K > D$) and used at rate D at the same time. Stock builds up at $(K - D)$ during production time $t_1 = Q/K$, then falls at D.



$Q^* = \sqrt{[\,2DC_0 / (\,C_h(1 - D/K)\,)\,]}$ $\quad I_{max} = Q^*(1 - D/K)$ $\quad TC_{min} = \sqrt{[\,2DC_0C_h(1 - D/K)\,]}$

$t_1 = Q^*/K$ (production), cycle time $t = Q^*/D$, $t_2 = t - t_1$. As $K \to \infty$ the model becomes EOQ Model I.

Example 6: $D = 2400$, $K = 4800$, $C_0 = $ Rs 100, $C_h = $ Rs 6. $1 - D/K = 0.5$.

$Q^* = \sqrt{(2 \times 2400 \times 100/(6 \times 0.5))} = \sqrt{160000} =$ 400 units; $I_{max} = 400 \times 0.5 =$ 200;

$TC = \sqrt{(2 \times 2400 \times 100 \times 6 \times 0.5)} =$ Rs 1200; production time = 400/4800 = 1/12 yr (30 days); cycle = 1/6 yr (60 days).

Model with shortages and production: $Q^* = \sqrt{[2DC_0(C_h+C_s)/(C_hC_s(1-D/K))]}$ (extension for exam awareness).

10. Buffer Stock and Reorder Level

Real demand and lead time vary, so a **buffer (safety) stock** is held to reduce the chance of a stock-out.

Reorder level (ROL) = (average demand × lead time) + buffer stock

Buffer stock = (maximum lead-time demand) - (average lead-time demand)

ROL (max-based) = maximum usage rate × maximum lead time. Minimum level = ROL - (normal usage × normal lead time)

Maximum level = ROL + ROQ - (minimum usage × minimum lead time). Average stock = minimum level + ROQ/2

Example: normal use 20 units/day, lead time 10 days; maximum use 30/day, lead time up to 12 days. ROL = 30 × 12 = 360; average lead-time demand = 200, so buffer stock = 360 - 200 = 160 units. If lead-time demand is normal with mean μ_L and s.d. σ_L: buffer = $z\,\sigma_L$ for a service level ($z = 1.65$ for 95%).

11. Quantity Discount Model - Single Price Break

The supplier offers a **lower price p_2** if $Q \geq q_b$(break quantity), else price p_1. Holding cost is usually a percentage i of price ($C_h = i\,p$). Total cost $TC = Dp + DC_0/Q + i\,p\,Q/2$.

Step 1: compute EOQ at the lowest price p_2. If $EOQ_2 \geq q_b$: it is feasible and OPTIMAL (stop).
Step 2: otherwise compute EOQ_1 at p_1 (feasible if $< q_b$) and its total cost TC_1.
Step 3: compute TC at $Q = q_b$ with price p_2. Choose the smaller total cost.

Example 7(a): D = 1000, C_0 = Rs 100, i = 20%; price Rs 50 for Q < 100, Rs 45 for Q ≥ 100

EOQ at Rs 45: $C_h = 9$, $Q = \sqrt{(2 \times 1000 \times 100/9)} = 149.07 \geq 100$ -> feasible, so $Q^* = 149$ units; $TC = 45000 + 670.8 + 670.8 =$ Rs 46,341.6.

Example 7(b): D = 1000, C_0 = Rs 50, i = 20%; price Rs 20 for Q < 500, Rs 18 for Q ≥ 500

Option	Q	Price	$TC = Dp + DC_0/Q + C_hQ/2$
EOQ at Rs 18 ($C_h = 3.6$)	166.7 < 500	-	not feasible
EOQ at Rs 20 ($C_h = 4$)	158.1	20	20000 + 316.2 + 316.2 = 20,632.5
Break quantity	500	18	18000 + 100 + 900 = 19,000

Minimum cost is at Q = 500 (Rs 19,000), so accept the discount and order the break quantity of 500.

12. Selective Inventory Control

Treating all items alike is wasteful. **Selective control** classifies items so that management effort is proportional to importance.

Technique	Basis of classification	Categories
ABC	annual consumption value (Pareto 80:20)	A, B, C
VED	criticality of item to production	Vital, Essential, Desirable
FSN	rate of consumption / movement	Fast, Slow, Non-moving
HML	unit price	High, Medium, Low
SDE	difficulty of procurement	Scarce, Difficult, Easy
XYZ	value of stock held	X (high), Y, Z (low)

ABC analysis - procedure

- Find annual usage value = annual demand × unit price for every item; arrange in **descending** order
- Compute cumulative value % and cumulative item %; draw the cumulative (Lorenz-type) curve; fix A/B/C cut-offs

Class	% of items	% of value	Control
A	10-20%	70-80%	tight: close records, frequent review, small safety stock
B	~30%	15-20%	moderate: periodic review
C	50-60%	5-10%	loose: bulk orders, simple records, large stock

Example 8

Item	Value (Rs '000)	Cum. value %	Cum. items %	Class
I1	450	45.0	12.5	A
I2	280	73.0	25.0	A
I3	120	85.0	37.5	B
I4	60	91.0	50.0	B
I5	40	95.0	62.5	C
I6	25	97.5	75.0	C
I7	15	99.0	87.5	C
I8	10	100.0	100.0	C

Class A: I1, I2 (25% of items, 73% of value). Class B: I3, I4 (25% items, 18%). Class C: I5-I8 (50% items, only 9% of value).

Extra: single-period (newsboy) model - probabilistic demand

For perishable items bought once. Let C_u = profit lost per unit short (selling price - cost) and C_o = loss per unit left over (cost - salvage). Stock Q^* such that $\mathbf{P(demand \le Q^*) = C_u/(C_u + C_o)}$ (critical ratio).

Example: cost Rs 10, price Rs 25, salvage Rs 5. $C_u = 15$, $C_o = 5$, critical ratio = 15/20 = 0.75.

If demand is uniform between 100 and 200: $Q^* = 100 + 0.75 \times 100 = 175$ units.

Combining ABC with VED (control matrix)

	V (vital)	E (essential)	D (desirable)
A (high value)	tight control, small stock	moderate	very low stock
B	tight control	moderate	periodic review
C (low value)	keep large safety stock	bulk order	loose control

13. Applications, Limitations and Practice Problems

Applications of queuing theory

- Banks, supermarkets, petrol pumps, toll plazas: deciding the number of counters (servers)
- Hospitals and call centres: staffing; machine repair: number of repairmen; airports: runway and gate allocation; computer networks: buffer sizes

Limitations of the simple models

- Assume Poisson arrivals and exponential service with constant rates; real systems have peaks and non-exponential times
- Steady state may never be reached for short periods; balking, reneging and priorities complicate the maths - then **simulation** is used
- Other models (names only): (M/M/1):(N/FIFO) finite source, (M/G/1) Pollaczek-Khinchine, ($M/E_k/1$) Erlang service, priority queues, queues in series

Practice problems with answers

No.	Problem	Answer
1	M/M/1, λ = 4/hr, μ = 6/hr. Find P_0, L_s, L_q, W_s, W_q	1/3; 2; 1.333; 0.5 hr; 0.333 hr
2	M/M/1, λ = 3/hr, μ = 5/hr. L_s and P(4 or more in system)	1.5; $0.6^4 = 0.1296$
3	M/M/2, λ = 6/hr, μ = 4/hr each. Find P_0, L_q, W_q	1/7; 1.929; 0.321 hr
4	EOQ: D = 5000, C_0 = 200, C_h = 10. Q*, TC	447.2 units; Rs 4472
5	Same with shortage cost C_s = 40. Q*, S*, TC	500; 400 (backorder 100); Rs 4000
6	Production model, K = 10000, same data as 4. Q*, I_{max}, TC	632.5; 316.2; Rs 3162
7	Normal use 15/day, lead 8 days; max use 25/day, lead 10 days. ROL, buffer stock	250; 130 units

Check of problem 7: ROL = 25 × 10 = 250; average lead-time demand = 15 × 8 = 120; buffer = 250 - 120 = 130.

Formula flash cards

Queue ρ: M/M/1 $\rho = \lambda/\mu$; M/M/c $\rho = \lambda/(c\mu)$. *Always* check $\rho < 1$ before using steady-state formulas.

Time conversion: W in hours × 60 = minutes. $P(n \geq k) = \rho^k$ for M/M/1 infinite. $P(\text{wait} > 0) = 1 - P_0 = \rho$ for M/M/1.

Inventory: ordering cost = DC_0/Q, holding = $C_hQ/2$, equal at Q*. Reorder level = lead-time demand + buffer. ABC: A ≈ 70-80% value.

14. Quick Revision & Important Questions

Queues: $M/M/1$: $L_s = \lambda/(\mu-\lambda)$, $W_s = 1/(\mu-\lambda)$, $L_q = \lambda^2/\mu(\mu-\lambda)$; finite: $\lambda' = \lambda(1-P_N)$; $M/M/c$: P_0 formula, L_q formula; Little: $L = \lambda W$.
Inventory: $Q^* = \sqrt{(2DC_0/C_h)}$; with shortages multiply by $\sqrt{((C_h+C_s)/C_s)}$; production: divide inside root by $(1 - D/K)$; discount: compare total costs.

Likely questions: (1) Explain elements of a queuing system; (2) Derive P_n, L_s, W_s for $M/M/1$; (3) Solve $M/M/1$ and $M/M/c$ numericals; (4) Derive EOQ with and without shortages; (5) Production lot-size model; (6) Quantity-discount problem; (7) ABC analysis; (8) Buffer stock.